Enterprise Multi-Floor Hotel Network Design and Implementation.

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CCNP-Level Enterprise Infrastructure Project (EVE-NG)

Project Title

Design and Implementation of a Secure Enterprise Multi-Floor Hotel Network with VLAN Segmentation, OSPF Routing, DHCP Services, SSH Management, ACL Enforcement, Wireless Guest Services and Layer-2 Security Controls.

Business Scenario

Vic Modern Hotel is launching a new branch in Bangalore. The hotel consists of three floors containing operational, administrative and technical departments. The organization requires a scalable, resilient and secure network infrastructure supporting both wired and wireless users while maintaining strict access control between departments and guest networks.

Project Objectives

- Design a hierarchical enterprise network.
- Implement VLAN-based segmentation.
- Configure Router-on-a-Stick Inter-VLAN Routing.
- Deploy DHCP services for all departments.
- Deploy dedicated guest wireless networks.
- Implement OSPF dynamic routing.
- Configure SSH for secure management.
- Implement Port Security.
- Enforce Guest Access Policies using ACLs.
- Verify enterprise connectivity and resiliency.

Network Architecture

Core Layer: • One Core Switch Distribution Layer:

- Floor-1 Distribution Switch
- Floor-2 Distribution Switch
- Floor-3 Distribution Switch Routing Layer:
- Floor-1 Router • Floor-2 Router
- Floor-3 Router Wireless Layer:
- Wireless Gateway Router – Floor 1
- Wireless Gateway Router – Floor 2
- Wireless Gateway Router – Floor 3

Department VLAN Design

VLAN 10 – IT – 192.168.1.0/24

VLAN 20 – Administration – 192.168.2.0/24

VLAN 30 – Sales – 192.168.3.0/24

VLAN 40 – Human Resources – 192.168.4.0/24

VLAN 50 – Finance – 192.168.5.0/24

VLAN 60 – Logistics – 192.168.6.0/24

VLAN 70 – Store – 192.168.7.0/24

VLAN 80 – Reception – 192.168.8.0/24

Guest Network Design

Floor-1 Guest Network – 192.168.100.0/24

Floor-2 Guest Network – 192.168.110.0/24

Floor-3 Guest Network – 192.168.120.0/24

Wireless Gateway Routers must: • Provide DHCP services

• Participate in OSPF

• Advertise guest routes

• Support controlled access policies

Router Interconnection Plan

F1 ↔ F2 : 10.10.10.0/30

F1 ↔ F3 : 10.10.10.4/30

F2 ↔ F3 : 10.10.10.8/30 . The triangular topology must provide route redundancy and OSPF failover.

Task 1 – Enterprise Topology Deployment

Build the complete topology in EVE-NG and document all interface assignments.

Task 2 – VLAN Configuration

Create all VLANs, assign switch ports and verify VLAN membership.

Task 3 – Trunk Configuration

Configure IEEE 802.1Q trunk links between Core and Distribution layers and toward Router-on-a-Stick links.

Task 4 – Inter-VLAN Routing

Configure Router-on-a-Stick using subinterfaces and verify communication between all authorized VLANs.

Task 5 – Department DHCP Services

Configure floor routers as DHCP servers for departmental VLANs. Exclude gateway addresses and verify lease allocation.

Task 6 – Guest DHCP Services

Configure Wireless Gateway Routers as DHCP servers for guest networks and verify dynamic addressing.

Task 7 – OSPF Deployment

Configure OSPF Process ID 1. Advertise: • VLAN Networks • Guest Networks • Router Interconnection Links Verify: show ip ospf neighbor show ip route ospf show ip protocols

Task 8 – OSPF Optimization

Configure passive interfaces on user-facing networks.

Students Must Explain:

- OSPF Neighbor Formation
- SPF Algorithm
- OSPF Convergence • Benefits of Passive Interfaces

Task 9 – Secure SSH Management

Configure SSH Version 2 on all routers and wireless gateway routers.

Requirements: • Generate RSA Keys

- Create Local Administrator Account
- Disable Telnet • Configure Login Local Authentication SSH Management Policy:
- Only TEST-PC in VLAN 10 may manage infrastructure devices.
- SSH from Guest Networks is prohibited.
- SSH from all non-IT VLANs is prohibited. Verification:
- Successful SSH from TEST-PC • Failed SSH from Guest Network
- Failed SSH from Non-IT VLAN

Task 10 – Port Security

Deploy TEST-PC in the IT Department. Requirements:

• Sticky MAC Learning • Maximum MAC Address = 1

• Violation Mode = Shutdown Verification: show port-security show port-security interface

Task 11 – Guest Access Control Policy

Implement ACLs enforcing the following policy.

PERMIT: • Guest Networks -> VLAN 10 (IT)

- Guest Networks -> VLAN 70 (Store)
- Guest Networks -> VLAN 80 (Reception)
- Guest Networks -> Other Guest Networks

DENY: • Guest Networks -> VLAN 20 (Administration)

- Guest Networks -> VLAN 30 (Sales)
- Guest Networks -> VLAN 40 (HR)
- Guest Networks -> VLAN 50 (Finance)
- Guest Networks -> VLAN 60 (Logistics)
- Guest Networks -> VLAN 80 (Reception)

Students must demonstrate both successful and blocked traffic.

Task 12 – Enterprise Verification

Verify:

• VLAN Connectivity

• Inter-VLAN Routing

• DHCP Allocation

• OSPF Route Learning

• OSPF Neighbor Relationships

• SSH Access Control

• ACL Enforcement

• Guest-to-Guest Communication

• Guest-to-IT Communication

• Guest-to-Store Communication

• Guest-to-Reception Communication

• End-to-End Connectivity

Task 13 – Failover Testing

All devices in each floor in every vlan must ping or reach the INTERNET. Guest devices in guest networks should reach the Internet as well. Make sure to make policies in the FORTIGATE Firewall to allow traffic only from LAN -> WAN. No one from the WAN side should by default reach LAN sites.

Task 14 – Internet WAN Connectivity

Simulate router link failures. • Observe OSPF reconvergence • Verify alternate path selection • Verify connectivity restoration

Bonus Enterprise Tasks:

• PortFast • BPDU Guard • Disable Unused Ports • Loopback Interfaces • OSPF Advertisement of Loopbacks • Infrastructure Hardening • Troubleshooting Documentation

Project Deliverables

1. Topology Diagram
2. VLAN Design Plan
3. IP Addressing Plan
4. OSPF Verification Output
5. DHCP Verification Output
6. SSH Verification Output
7. ACL Verification Output
8. Port Security Output
9. Connectivity Test Results
10. Final Project Documentation